

DD_ASR-01 - Technical Trouble

Version: 1.0 Status: Draft for review Bundle: 10_other Audience: customer care developers, ticketing developers, work-order developers, NOC, QA Related DDs:

- DD_TCK-01 - Ticketing and Case Management
- DD_WO-01 - Work Order Module
- DD_WO-01-FLOW-SUPPORT - Support Work Order Flow
- DD_PROV-INT-01 - Provisioning Broadcast and Reconciliation
- DD_CUST-INT-01 - Customer Interaction Timeline
- DD_BIL-02-ADJ-01 - Invoice Adjustments
- FE-APP-01 - BSS Backoffice Web App
- FE-APP-04 - Customer Self Care Frontoffice App

MVP simplification baseline: ASR-01 is implemented as a TCK-01 ticket type, ruleset, SLA profile, and optional workflow satellite. It is not a separate ticketing service, database, or deployment in the first release.

1. Purpose

ASR-01 defines the technical trouble process for customer-reported service faults.

Examples:

- no internet;
- slow internet;
- intermittent connection;
- no voice service;
- TV/IPTV not working;
- equipment suspected faulty;
- repeated technical issue after recent work order;
- area outage affecting multiple customers.

This DD does not create a new ticketing engine. Technical trouble is implemented as a configured TECHNICAL_SUPPORT ticket type in TCK-01, with ASR-specific category rules, diagnostics, work-order triggers, outage handling, and optional billing-credit hooks.

2. Ownership Boundary

Concern	Owner
Ticket lifecycle, assignment, comments, attachments, timeline	TCK-01
Technical trouble category rules and diagnostic flow	ASR-01 configuration and logic inside TCK/ASR handlers
Support work order execution	WO-01 / WO support flow
Field notes, site visit, technician findings	WO-01
Network command/reconciliation visibility	PROV-INT-01 read APIs
Subscription state	SUB-LM / SUB-WF
Customer/account data	ILM-CFG-01
Credit note or compensation for outage	BIL-02-ADJ-01, optionally triggered from ASR resolution
Customer timeline projection	CUST-INT-01

Rule: ASR-01 does not own a separate case table. The ticket record remains in TCK-01.

3. When To Use This Flow

Use ASR-01 when the customer issue is a suspected service fault that may require technical diagnosis or field work.

Use other flows when:

Scenario	Use
Customer asks for package/pricing information	ASR-02 Information Request
Customer complains about service quality, staff conduct, billing fairness, or unresolved issue	ASR-03 Complaint
Customer asks for an operational action such as SIP reset, address change, or service parameter change	ASR-04 Service Request
Customer disputes an invoice amount	TCK billing complaint plus BIL adjustment flow

4. High Level Flow

```
flowchart TD
    A["Customer reports issue"]
    B["Create TCK ticket<br/>type TECHNICAL_SUPPORT"]
    C["Run ASR diagnostic checklist"]
    D{"Known outage?"}
    E["Attach outage reference<br/>notify customer"]
    F{"Remote fix possible?"}
    G["Run remote action / guide customer"]
    H{"Resolved?"}
    I["Create support WO"]
    J["Wait for WO finalization"]
    K["Review findings"]
    L{"Credit needed?"}
    M["Start BIL adjustment request"]
    N["Resolve ticket"]

    A --> B --> C --> D
    D -->|Yes| E --> N
    D -->|No| F
    F -->|Yes| G --> H
    H -->|Yes| N
    H -->|No| I
    I --> J --> K --> L
    L -->|Yes| M --> N
    L -->|No| N
```

5. Ticket Categories

ASR-01 uses TCK-01.ticket.type_code = TECHNICAL_SUPPORT.

Recommended initial WIK categories:

Category code	Meaning	WO allowed	Default priority
NO_INTERNET	Customer has no data connectivity.	Yes	P2
SLOW_SPEED	Customer reports speed below subscribed profile.	Yes	P3
INTERMITTENT_CONNECTION	Connection drops repeatedly.	Yes	P3
NO_VOICE	Voice service not working.	Yes	P2
NO_TV	TV/IPTV service not working.	Yes	P3
CPE_FAULT_SUSPECTED	ONT/router/ATA/STB suspected faulty.	Yes	P3
AREA_OUTAGE	Customer issue linked to known area outage.	No by default	P1
REPEAT_FAULT	Same issue reopened or repeated after recent resolution.	Yes	P2

These categories are stored in TCK category configuration. ASR-01 adds rules on top of those categories.

6. ASR-Specific Tables

ASR-01 can be implemented with a small set of ASR configuration tables inside the ticketing schema. Ticket records still use TCK tables.

6.1 asr_trouble_category_config

Stores technical-trouble behavior per ticket category.

Use this table when deciding required diagnostics, WO policy, and resolution behavior.

Column	Type	Purpose	Sample
config_id	UUID PK	Config id.	asr-cat-001
operator_code	TEXT	Operator/tenant.	WIK
category_code	TEXT	TCK category.	NO_INTERNET
service_class_code	TEXT	INTERNET, VOICE, TV, ANY.	INTERNET
diagnostic_profile_code	TEXT	Checklist profile.	INET_BASIC_DIAG
wo_policy	TEXT	NONE, OPTIONAL, REQUIRED_AFTER_DIAG, AUTO_CREATE.	REQUIRED_AFTER_DIAG
default_wo_kind	TEXT	WO kind if created.	SUPPORT
repeat_fault_window_days	INTEGER	Window for repeat-fault detection.	30
credit_policy_code	TEXT NULL	Optional credit policy.	OUTAGE_CREDIT_STANDARD
status	TEXT	ACTIVE, RETIRED.	ACTIVE

Recommended indexes:

- Unique (operator_code, category_code).

6.2 asr_diagnostic_profile

Stores a diagnostic checklist definition.

Use this table to drive consistent customer-care diagnosis before creating a WO.

Column	Type	Purpose	Sample
profile_id	UUID PK	Profile id.	diag-001
operator_code	TEXT	Operator/tenant.	WIK
profile_code	TEXT	Stable profile code.	INET_BASIC_DIAG
display_name	TEXT	Backoffice label.	Internet Basic Diagnostics
required_checks_json	JSONB	Ordered checks.	["SUB_ACTIVE", "OUTAGE_CHECK", "ONT_STATUS", "LAST_PROV_COMMAND"]
status	TEXT	ACTIVE, RETIRED.	ACTIVE

6.3 asr_ticket_diagnostic_result

Stores diagnostic results captured for a ticket.

Use this table to avoid free-text-only troubleshooting.

Column	Type	Purpose	Sample
result_id	UUID PK	Result id.	diagres-001
ticket_id	TEXT	TCK ticket id.	tck-001
profile_code	TEXT	Diagnostic profile used.	INET_BASIC_DIAG
check_code	TEXT	Check performed.	ONT_STATUS
result_code	TEXT	PASS, FAIL, UNKNOWN, SKIPPED.	FAIL
result_payload_json	JSONB	Structured result.	{"ontOnline":false}
captured_by_user_id	TEXT	User/system.	usr-care-001
captured_at	TIMESTAMP	Capture time.	2026-06-03T09:10:00Z

Recommended indexes:

- (ticket_id, check_code).

6.4 asr_outage_link

Stores a ticket link to a known outage or network incident.

Use this table when multiple customer tickets are caused by the same area/platform incident.

Column	Type	Purpose	Sample
outage_link_id	UUID PK	Link id.	aol-001
ticket_id	TEXT	TCK ticket id.	tck-001
outage_ref	TEXT	External/NOC outage id.	OUT-NRB-20260603-01
outage_status	TEXT	OPEN, RESOLVED, CANCELLED.	OPEN
linked_by_user_id	TEXT	User/system.	usr-noc-001
linked_at	TIMESTAMP	Link time.	2026-06-03T09:12:00Z

6.5 asr_compensation_recommendation

Stores optional compensation recommendation after verified trouble/outage.

Use this table when ASR recommends credit but BIL-02-ADJ-01 owns the actual adjustment.

Column	Type	Purpose	Sample
recommendation_id	UUID PK	Recommendation id.	acr-001
ticket_id	TEXT	TCK ticket id.	tck-001
operator_code	TEXT	Operator/tenant.	WIK
reason_code	TEXT	Compensation reason.	SERVICE_OUTAGE_VERIFIED
recommended_amount	DECIMAL(18,2) NULL	Suggested credit amount.	250.00
currency_code	TEXT	Currency.	KES
adjustment_request_id	TEXT NULL	BIL adjustment request if created.	adj-001
status	TEXT	RECOMMENDED, REQUESTED, APPROVED, REJECTED, CANCELLED.	RECOMMENDED
created_at	TIMESTAMP	Creation timestamp.	2026-06-03T11:00:00Z

7. APIs

ASR-01 primarily uses TCK APIs. These endpoints may be implemented as ASR-specific commands within the TCK module.

7.1 Create Technical Trouble Ticket

```
POST /api/tickets
Idempotency-Key: care-nointernet-20260603-001
Content-Type: application/json

{
  "operatorCode": "WIK",
  "typeCode": "TECHNICAL_SUPPORT",
  "categoryCode": "NO_INTERNET",
  "sourceChannel": "CALL_CENTER",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "subject": "Customer reports no internet",
  "description": "Customer says ONT has LOS light blinking after overnight rain.",
  "customerVisible": true
}
```

7.2 Run Diagnostic

```
POST /api/asr/technical-trouble/{ticket_id}/diagnostics
Content-Type: application/json

{
  "profileCode": "INET_BASIC_DIAG",
```

```

"results": [
  {
    "checkCode": "SUB_ACTIVE",
    "resultCode": "PASS",
    "payload": {"subscriptionStatus": "ACTIVE"}
  },
  {
    "checkCode": "OUTAGE_CHECK",
    "resultCode": "PASS",
    "payload": {"knownOutage": false}
  },
  {
    "checkCode": "ONT_STATUS",
    "resultCode": "FAIL",
    "payload": {"ontOnline": false, "lastSeenAt": "2026-06-02T23:10:00Z"}
  }
]
}

```

7.3 Create Support Work Order From Ticket

Use TCK's create-WO command, with ASR context in payload:

```

POST /api/tickets/{ticket_id}/work-orders
Idempotency-Key: tck-001-support-wo-001
Content-Type: application/json

{
  "workOrderKind": "SUPPORT",
  "reasonCode": "NO_INTERNET",
  "priorityCode": "P2",
  "homepassId": "HP-300045",
  "subscriptionId": "SUB-700045",
  "diagnosticSummary": {
    "profileCode": "INET_BASIC_DIAG",
    "failedChecks": ["ONT_STATUS"]
  },
  "customerReportedSymptoms": ["LOS_BLINKING", "NO_CONNECTIVITY"]
}

```

7.4 Recommend Compensation

```

POST /api/asr/technical-trouble/{ticket_id}/compensation-recommendations
Content-Type: application/json

{
  "reasonCode": "SERVICE_OUTAGE_VERIFIED",
  "recommendedAmount": 250.00,
  "currency": "KES",
  "notes": "Verified outage lasted 10 hours and affected subscribed internet service."
}

```

If operations approves creating a formal credit request, ASR calls BIL-02-ADJ-01. ASR does not issue the credit note directly.

8. Events

Event	Topic	When emitted
TechnicalTroubleTicketCreated	sophix.asr.technical-trouble.created	Technical-support ticket created.
TechnicalTroubleDiagnosticCompleted	sophix.asr.technical-trouble.diagnosticcompleted	Diagnostic profile results captured.
TechnicalTroubleLinkedToOutage	sophix.asr.technical-trouble.outagelinked	Ticket linked to outage.
TechnicalTroubleWorkOrderRequested	sophix.asr.technical-trouble.workorderrequested	TCK creates a support WO due to ASR flow.
TechnicalTroubleResolved	sophix.asr.technical-trouble.resolved	Ticket resolved through remote fix, outage closure, or WO result.
TechnicalTroubleCompensationRecommended	sophix.asr.technical-trouble.compensationrecommended	ASR recommends a BIL adjustment.

TCK still emits its generic ticket events. ASR events are business-specific projections for reporting and downstream modules.

9. Work Order Rules

ASR may request a support WO only when:

- ticket type is TECHNICAL_SUPPORT;
- category config has wo_policy != NONE;
- required diagnostics are captured unless category has AUTO_CREATE;
- no active linked support WO already exists, unless repeat WO is explicitly allowed;
- customer/subscription/HomePass references are present where required;
- user has ticket.workorder.create or equivalent permission from EM-CFG-03.

After WO finalization:

WO result	Ticket action
Fixed/resolved	Move ticket to UNDER_REVIEW or RESOLVED based on category policy.
Customer unavailable	Move ticket to WAITING_CUSTOMER.
Network construction needed	Keep WAITING_INTERNAL and optionally create escalation ticket/WO.
No fault found	Move UNDER_REVIEW; customer care decides closure.
Repeat fault flagged	Reopen/escalate as REPEAT_FAULT.

10. Compensation Rules

Compensation is optional and policy-driven.

ASR may recommend compensation when:

- trouble is verified as operator/network fault;
- service impact duration exceeds configured threshold;
- issue affected a billable active subscription;
- ticket/WO evidence supports the claim;
- no previous compensation was granted for the same outage period.

ASR calls BIL-02-ADJ-01 only to create an adjustment request. BIL owns approval and credit-note generation.

Example adjustment request context:

```
{
  "sourceModule": "ASR-01",
  "sourceRefId": "tck-001",
  "customerId": "CUS-100045",
  "subscriptionId": "SUB-700045",
  "adjustmentKind": "CREDIT_NOTE",
  "reasonCode": "SERVICE_OUTAGE_VERIFIED",
  "amount": 250.00,
  "currency": "KES",
  "scope": "AMOUNT",
  "notes": "ASR verified 10-hour internet outage."
}
```

11. Frontend Usage

11.1 FE-APP-01 Backoffice

Backoffice uses ASR-01/TCK APIs for:

- Create technical trouble ticket from customer 360.
- Guided diagnostic checklist.
- Known outage lookup/link.
- Create support WO.

- View WO progress on ticket.
- Resolve ticket with structured resolution code.
- Recommend compensation.
- View repeat fault history.

11.2 FE-APP-04 Customer Self-Care

Self-care can expose a simplified technical support request:

- choose affected service;
- choose symptom;
- attach photo if relevant;
- create TCK ticket with `sourceChannel = SELF_CARE`;
- show customer-safe ticket status and WO appointment where applicable.

Self-care must not expose internal diagnostic payloads, NOC notes, or technician private notes.

12. Infra Recommendation

ASR-01 should run inside the TCK service/module. No separate ASR pod is required for medium budget deployment.

Recommended deployment:

Component	Recommendation
API	Existing TCK API pods.
DB	Existing TCK schema with ASR config tables.
Workers	Existing TCK workers for ticket events and WO event consumers.
Cache	Optional cache for category/diagnostic config.

AWS deployment:

- Same ECS/EKS service as TCK.
- Same RDS schema as TCK.
- Same Kafka/MSK event infrastructure.
- CloudWatch alarms for diagnostic failures, WO creation failures, and ASR SLA breach counts.

13. Testing Requirements

13.1 Unit Tests

- Technical trouble category config resolves diagnostic profile.
- Required diagnostic missing blocks WO creation.
- Known outage link moves ticket to correct status.
- Repeat fault detection identifies prior resolved ticket within window.
- Compensation recommendation does not create credit directly.

13.2 Integration Tests

- Create `NO_INTERNET` ticket.
- Capture diagnostic results.
- Create support WO through TCK.
- Consume `WorkOrderFinalized` and update ticket.

- Create compensation recommendation and BIL adjustment request.

13.3 End To End Tests

- Call center creates no-internet ticket, diagnostic fails ONT check, WO is created, technician fixes issue, ticket resolves.
- Customer creates self-care trouble ticket, care team links it to known outage, customer sees outage-safe status.
- Repeat fault within 30 days is escalated as REPEAT_FAULT.

14. Definition Of Done

ASR-01 is complete when:

- Technical trouble categories are seeded in TCK.
- ASR category config and diagnostic profiles exist.
- Diagnostic capture APIs work.
- WO creation is gated by ASR rules.
- WO events update the ticket timeline and status.
- Known outage linking works.
- Compensation recommendation can create a BIL adjustment request without directly crediting customer.
- Customer timeline receives ASR/TCK events.
- Backoffice can execute the full trouble flow.